

Easy (10 MCQs)

1.

What protocol does PPP use to establish and configure a link?

- A) IP
- B) LCP
- C) ARP
- D) DHCP

Answer: B) LCP

Explanation: PPP uses the Link Control Protocol (LCP) to establish, configure, and test the data link connection.

2.

Which authentication methods are supported by PPP?

- A) WEP and WPA
- B) PAP and CHAP
- C) SSL and TLS
- D) SSH and Telnet

Answer: B) PAP and CHAP

Explanation: PPP supports PAP (Password Authentication Protocol) and CHAP (Challenge Handshake Authentication Protocol) for authentication.

3.

What does MLPPP stand for?

- A) Multiple Link PPP
- B) Multilink PPP

- C) Multi-layer PPP
- D) Multiple Layer PPP

Answer: B) Multilink PPP

Explanation: MLPPP stands for Multilink Point-to-Point Protocol, which allows multiple physical links to be combined logically.

4.

PPPoE encapsulates PPP frames inside what kind of frames?

- A) IP frames
- B) Ethernet frames
- C) UDP frames
- D) TCP frames

Answer: B) Ethernet frames

Explanation: PPPoE encapsulates PPP frames within Ethernet frames for broadband connections.

5.

What command is used to verify PPP authentication sessions?

- A) show ip interface
- B) show authentication sessions
- C) show interfaces status
- D) show pppoe sessions

Answer: B) show authentication sessions

Explanation: This command shows active authentication sessions on the router.

6.

GRE tunnels encapsulate packets at which layer?

- A) Data Link
- B) Network
- C) Transport
- D) Session

Answer: B) Network

Explanation: GRE operates at Layer 3 (Network layer) and encapsulates Layer 3 packets inside IP packets.

7.

Which BGP command is used to advertise connected networks?

- A) neighbor
- B) network
- C) redistribute
- D) route-map

Answer: B) network

Explanation: The network command in BGP advertises connected networks into BGP.

8.

What is the default encapsulation for PPP on a serial interface?

- A) HDLC
- B) PPP
- C) Frame Relay
- D) Ethernet

Answer: A) HDLC

Explanation: By default, Cisco uses HDLC encapsulation on serial interfaces; PPP must be explicitly configured.

9.

Which authentication protocol is more secure for PPP?

- A) PAP
- B) CHAP
- C) WEP
- D) None

Answer: B) CHAP

Explanation: CHAP is more secure than PAP because it uses a challenge-response mechanism.

10.

What IP address configuration is typically assigned to GRE tunnel interfaces?

- A) Public IP addresses
- B) Loopback IP addresses
- C) Private IP addresses
- D) Any IP address within the tunnel subnet

Answer: D) Any IP address within the tunnel subnet

Explanation: GRE tunnel interfaces are assigned IPs from a subnet that is unique to the tunnel, typically private addresses.

Medium (30 MCQs)

11.

Which of the following is true about MLPPP?

- A) It requires only one physical link
- B) It can combine multiple physical links into one logical link
- C) It does not support authentication
- D) It operates only on Ethernet interfaces

Answer: B) It can combine multiple physical links into one logical link

Explanation: MLPPP allows multiple physical serial links to be bonded logically, increasing bandwidth.

12.

In PPPoE configuration, which virtual interface represents the PPP session?

- A) Dialer interface
- B) Loopback interface
- C) Tunnel interface
- D) Multilink interface

Answer: A) Dialer interface

Explanation: PPPoE client-side is typically configured with a dialer interface that represents the PPP session.

13.

What is the purpose of the debug ppp negotiation command?

- A) To show interface status
- B) To troubleshoot PPP connection and authentication steps
- C) To configure PPP settings
- D) To reset PPP sessions

Answer: B) To troubleshoot PPP connection and authentication steps

Explanation: This debug command helps identify where PPP negotiation or authentication is failing.

14.

Which protocol encapsulates GRE packets?

- A) TCP
- B) UDP
- C) IP
- D) ICMP

Answer: C) IP

Explanation: GRE packets are encapsulated inside IP packets.

15.

In eBGP, what is the significance of the AS number?

- A) It identifies the router type
- B) It identifies the autonomous system to which a router belongs
- C) It configures routing protocol version
- D) It sets the router's IP address

Answer: B) It identifies the autonomous system to which a router belongs

Explanation: AS numbers are used to identify autonomous systems in BGP.

16.

Which of the following commands creates a BGP neighbor relationship?

- A) router bgp <AS> neighbor <IP> remote-as <AS>
- B) neighbor <IP> enable bgp
- C) router ospf neighbor <IP>
- D) bgp neighbor enable <IP>

Answer: A) router bgp <AS> neighbor <IP> remote-as <AS>

Explanation: This command defines a BGP neighbor and the remote AS for eBGP.

17.

Which of the following is NOT true about PPPoE?

- A) It allows Ethernet clients to connect using PPP authentication
- B) It encapsulates IP packets inside Ethernet frames
- C) It requires virtual-template interfaces
- D) It is used to connect two routers via serial link only

Answer: D) It is used to connect two routers via serial link only

Explanation: PPPoE runs over Ethernet, not serial links.

18.

What is the primary purpose of the network command in BGP?

- A) To enable BGP on an interface
- B) To advertise networks into BGP routing updates
- C) To configure static routes
- D) To create neighbor relationships

Answer: B) To advertise networks into BGP routing updates

Explanation: The network command tells BGP which networks to advertise.

19.

Which of these statements is true regarding GRE tunnels?

- A) GRE can only encapsulate IPv4 packets
- B) GRE does not provide encryption by default
- C) GRE tunnels cannot be used over the public internet
- D) GRE provides built-in authentication

Answer: B) GRE does not provide encryption by default

Explanation: GRE provides encapsulation but no encryption; encryption must be added via IPSec.

20.

Which of the following is a correct step in troubleshooting PPPoE?

- A) Check the show pppoe session command output
- B) Check VLAN assignments on tunnel interfaces
- C) Disable authentication on the dialer interface
- D) Assign static IP addresses on dialer interfaces

Answer: A) Check the show pppoe session command output

Explanation: This command helps verify active PPPoE sessions.

21.

How do you verify BGP neighbor status?

- A) show ip bgp neighbors

- B) show ip bgp summary
- C) show bgp interfaces
- D) show ip route

Answer: B) show ip bgp summary

Explanation: This command displays neighbor states and other BGP summary info.

22.

Which PPP authentication protocol sends credentials in plaintext?

- A) CHAP
- B) PAP
- C) MS-CHAP
- D) EAP

Answer: B) PAP

Explanation: PAP sends username and password in plaintext; CHAP is more secure.

23.

Which command verifies the IP address on a GRE tunnel interface?

- A) show ip interface brief
- B) show ip route tunnel
- C) show interfaces tunnel
- D) show tunnel ip address

Answer: A) show ip interface brief

Explanation: This shows IP addresses and statuses of all interfaces including tunnels.

24.

What is the function of the local authentication in PPP?

- A) To authenticate users with an external server
- B) To authenticate users with a locally configured username and password
- C) To disable authentication on the interface
- D) To authenticate users through a third-party device

Answer: B) To authenticate users with a locally configured username and password

Explanation: Local authentication uses usernames and passwords configured locally on the router.

25.

What is the role of the virtual-template interface in PPPoE?

- A) To define physical interface parameters
- B) To serve as a template for dialer interfaces
- C) To create a physical PPP interface
- D) To act as a backup interface

Answer: B) To serve as a template for dialer interfaces

Explanation: Virtual-template interfaces provide a template for dynamically created dialer interfaces in PPPoE.

26.

In eBGP, the "next-hop" attribute usually refers to:

- A) The neighbor's loopback address
- B) The IP address of the router advertising the route

- C) The IP address of the router receiving the route
- D) The default gateway

Answer: B) The IP address of the router advertising the route

Explanation: The next-hop is the IP address the router should forward traffic to reach the destination network.

27.

Which of the following commands binds physical interfaces to an MLPPP interface?

- A) interface multilink 1
- B) multilink group 1
- C) channel-group 1 mode active
- D) multilink bundle 1

Answer: B) multilink group 1

Explanation: Physical interfaces are bound to a multilink group with multilink group <number>.

28.

Which command is used to debug BGP?

- A) debug ip bgp
- B) debug bgp route
- C) debug ip routing
- D) debug bgp neighbor

Answer: A) debug ip bgp

Explanation: debug ip bgp provides detailed info on BGP events for troubleshooting.

29.

What does the GRE tunnel source command specify?

- A) The IP address where GRE traffic originates
- B) The IP address where GRE traffic terminates
- C) The IP address of the remote tunnel endpoint
- D) The IP address assigned to the GRE tunnel interface

Answer: A) The IP address where GRE traffic originates

Explanation: The source IP is the local IP from which GRE packets are sent.

30.

When configuring eBGP, which command enables advertisement of a network?

- A) router bgp <AS>
- B) network <network> mask <mask>
- C) neighbor <IP> remote-as <AS>
- D) advertise network <network>

Answer: B) network <network> mask <mask>

Explanation: This command tells BGP to advertise the specified network.

Hard (10 MCQs)

31.

Which PPP frame is used to negotiate authentication options?

- A) LCP Configure-Request
- B) PAP Authenticate-Request
- C) CHAP Challenge
- D) IPCP Configure-Request

Answer: A) LCP Configure-Request

Explanation: LCP Configure-Request frames negotiate link parameters including authentication protocols.

32.

Which of the following is true about PPPoE intermediate devices?

- A) They terminate PPP sessions
- B) They act as Layer 2 bridges forwarding PPPoE frames
- C) They encrypt PPP frames
- D) They authenticate PPP sessions

Answer: B) They act as Layer 2 bridges forwarding PPPoE frames

Explanation: Intermediate devices in PPPoE act as Layer 2 bridges, not terminating sessions.

33.

In MLPPP, which mechanism ensures packet order delivery?

- A) Multilink sequence numbers
- B) Time stamping
- C) LCP negotiation
- D) IP fragmentation

Answer: A) Multilink sequence numbers

Explanation: MLPPP uses sequence numbers to reorder packets arriving out of order.

34.

What is the default administrative distance of eBGP routes?

- A) 90
- B) 110
- C) 20
- D) 200

Answer: C) 20

Explanation: eBGP routes have an administrative distance of 20, making them more trusted than iBGP.

35.

How does eBGP handle next-hop addresses in route advertisement?

- A) It modifies next-hop to its own IP address
- B) It preserves the original next-hop address
- C) It sets next-hop to 0.0.0.0
- D) It ignores next-hop attribute

Answer: A) It modifies next-hop to its own IP address

Explanation: eBGP sets the next-hop attribute to its own IP before advertising to neighbors.

36.

What is the significance of the configuration register value 0x2102 in Cisco routers?

- A) Boot normally from IOS stored in flash
- B) Enter ROMMON mode
- C) Boot from TFTP server
- D) Disable password checking

Answer: A) Boot normally from IOS stored in flash

Explanation: 0x2102 tells the router to load the IOS from flash on boot.

37.

Which command allows you to test the path a packet takes using APIC-EM Path Trace?

- A) show path-trace
- B) apic-em path-trace
- C) path-trace trace
- D) apic-em path-trace acl

Answer: D) apic-em path-trace acl

Explanation: This command verifies ACL impact along the path.

38.

What happens if the BGP network command advertises a network not in the routing table?

- A) Network is advertised anyway
- B) Network is ignored
- C) BGP crashes
- D) Router generates an error message

Answer: B) Network is ignored

Explanation: BGP only advertises networks present in the routing table.

39.

In a GRE tunnel, if the tunnel interface goes down, what might be a cause?

- A) Incorrect source or destination IP in tunnel config
- B) Missing VLAN tag
- C) Disabled routing protocol on the tunnel
- D) PPP authentication failure

Answer: A) Incorrect source or destination IP in tunnel config

Explanation: Tunnel status depends on correct endpoint IP addresses.

40.

Which command is used to set PPP authentication to CHAP on a serial interface?

- A) ppp authentication chap
- B) ppp enable chap
- C) authentication ppp chap
- D) enable ppp chap

Answer: A) ppp authentication chap

Explanation: This command enables CHAP authentication on the interface.

41.

Which feature does PPPoE client use to maintain the session over Ethernet?

- A) Session ID
- B) VLAN tagging
- C) IP fragmentation
- D) Route caching

Answer: A) Session ID

Explanation: PPPoE uses session IDs to maintain separate PPP sessions over Ethernet.

42.

Which of the following commands configures a GRE tunnel source as a loopback interface?

- A) tunnel source Loopback0
- B) tunnel source GigabitEthernet0/0
- C) source tunnel Loopback0
- D) ip tunnel source Loopback0

Answer: A) tunnel source Loopback0

Explanation: This sets the tunnel source IP to the IP address of Loopback0.

43.

When configuring MLPPP, which command sets the bandwidth on the multilink interface?

- A) bandwidth <kbps>
- B) multilink bandwidth <kbps>
- C) interface bandwidth <kbps>
- D) link bandwidth <kbps>

Answer: A) bandwidth <kbps>

Explanation: Bandwidth on multilink interfaces is set using the standard bandwidth command.

44.

What is the first step in establishing a PPP connection?

- A) IPCP negotiation
- B) Authentication
- C) LCP link establishment
- D) Data transfer

Answer: C) LCP link establishment

Explanation: PPP begins with LCP establishing and configuring the link.

45.

Which of the following is true for eBGP multihop?

- A) It is used when eBGP neighbors are not directly connected
- B) It disables authentication
- C) It requires VRF configuration
- D) It allows iBGP peers to communicate

Answer: A) It is used when eBGP neighbors are not directly connected

Explanation: eBGP multihop allows BGP peers to communicate over multiple hops.

46.

What command configures a local username for PPP CHAP authentication?

- A) username <name> password <pwd>

- B) local user <name> password <pwd>
- C) user <name> secret <pwd>
- D) enable username <name> password <pwd>

Answer: A) username <name> password <pwd>

Explanation: This command creates a local user for authentication.

47.

Which protocol is commonly paired with GRE to provide encryption?

- A) IPSec
- B) SSL
- C) TLS
- D) SSH

Answer: A) IPSec

Explanation: GRE tunnels are often combined with IPSec for secure encrypted transport.

48.

Which command verifies that the BGP session is established and the state is "Established"?

- A) show ip bgp neighbors | include State
- B) show ip bgp summary
- C) show bgp neighbor status
- D) show ip route bgp

Answer: B) show ip bgp summary

Explanation: This shows the session state as Established.

49.

What is the effect of the "no shutdown" command on a tunnel interface?

- A) Enables the tunnel interface
- B) Disables the tunnel interface
- C) Removes the tunnel interface
- D) Resets the tunnel interface

Answer: A) Enables the tunnel interface

Explanation: The "no shutdown" command activates the interface.

50.

What is the default TTL value used in GRE packets?

- A) 255
- B) 64
- C) 128
- D) 1

Answer: A) 255

Explanation: GRE packets use a default TTL of 255 in the outer IP header.